

MODEL QUESTION PAPER
PRODUCTION & OPERATIONS MANAGEMENT- SET I

(Maximum marks: 75)

(Time: 3 hours)

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 marks)

1	 production refers to the manufacturing of large volume of a single or a very few varieties of products with a standard set of processes.	M 1.04	R
2	In type of layout, machines and equipments arranged in the sequence or order in which they are used in the manufacture of a given product.	M 1.06	R
3	refers to the use of statistical methods in the monitoring and maintaining of the quality of products and services.	M 2.06	R
4	 is the prediction of sales volume that a company can estimate to achieve in specified period of time in future.	M 2.02	R
5	Under method of sales forecasting, seller sell his product in a part of the market for some time and make the assessment of the sales based on these test sales.	M 2.03	R
6	TQM stands for.....	M 3.09	R
7	purchasing refers to the purchase of materials by a single purchase department.	M 3.03	R
8	The is the longest sequence of tasks that must be completed to successfully conclude a project, from start to finish.	M 4.03	R
9	refers to any investment opportunity which is to be exploited for profit.	M 4.06	R

PART B

II Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24marks)

1	Explain the different production systems.	M 1.04	U
2	Describe the types of plant layouts.	M 1.06	U
3	Explain production planning and control.	M 2.01	U
4	List the objectives of quality control.	M 2.05	U
5	Elaborate aggregate planning.	M 2.09	U
6	How will you explain centralized purchasing?	M 3.04	U
7	Summarize value analysis.	M 3.06	U
8	State the objectives of network analysis.	M 4.01	R
9	Describe the characteristic of projects.	M 4.05	U
10	Construct the network diagram for the given project.	M 4.03	A

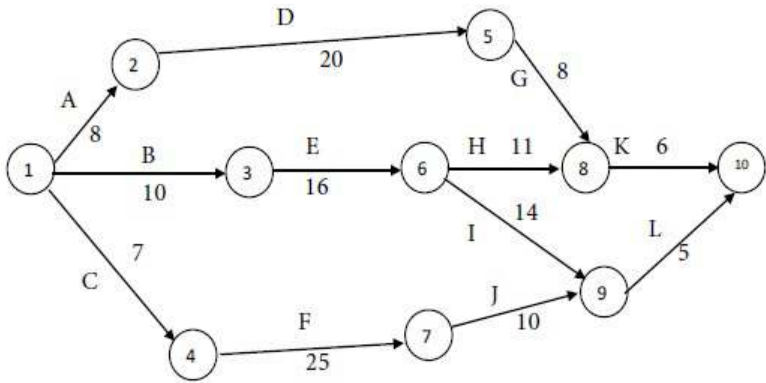
Activity	Predecessor Activity	Duration (Weeks)
A	-	3
B	A	5
C	A	7
D	B	10
E	C	5
F	D,E	4

PART C

Answer all questions. Each question carries seven marks

(6x7=42 marks)

III	Differentiate between Production and Operations management. OR	M 1.02	U																																													
IV	What are Process charts? Describe the different types of process charts.	M 1.08, 09	U																																													
V	Explain capacity planning.	M2.08	U																																													
VI	OR Outline the different techniques of sales forecasting	M2.03	U																																													
VII	Explain the methods of purchasing. OR	M3.03	U																																													
VIII	Differentiate centralized and decentralized purchasing.	M3.04	U																																													
IX	Identify the phases of project management. OR	M4.07	U																																													
X	A small project consisting of eight activities has the following characteristics: Time – Estimates (in weeks) <table><tr><th>Activity</th><th>Preceding activity</th><th>Most optimistic time (a)</th><th>Most likely time (m)</th><th>Most Pessimistic time (b)</th></tr><tr><td>A</td><td>None</td><td>2</td><td>4</td><td>12</td></tr><tr><td>B</td><td>None</td><td>10</td><td>12</td><td>26</td></tr><tr><td>C</td><td>A</td><td>8</td><td>9</td><td>10</td></tr><tr><td>D</td><td>A</td><td>10</td><td>15</td><td>20</td></tr><tr><td>E</td><td>A</td><td>7</td><td>7.5</td><td>11</td></tr><tr><td>F</td><td>B,C</td><td>9</td><td>9</td><td>9</td></tr><tr><td>G</td><td>D</td><td>3</td><td>3.5</td><td>7</td></tr><tr><td>H</td><td>E,F,G</td><td>5</td><td>5</td><td>5</td></tr></table>	Activity	Preceding activity	Most optimistic time (a)	Most likely time (m)	Most Pessimistic time (b)	A	None	2	4	12	B	None	10	12	26	C	A	8	9	10	D	A	10	15	20	E	A	7	7.5	11	F	B,C	9	9	9	G	D	3	3.5	7	H	E,F,G	5	5	5	M4.02	A
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F	B,C	9	9	9																																												
G	D	3	3.5	7																																												
H	E,F,G	5	5	5																																												

	(i) Draw the PERT network for the project. (ii) Prepare the activity schedule for the project. (iii) Determine the critical path. (iv) If a 30- week deadline is imposed, what is the probability that the project will be finished within the time limit?		
XI	Explain various modern production management tools. OR	M4.07	U
XII	What do you understand by codification? Explain in detail.	M4.02	A
XIII	Find out the completion time and the critical activities for the following project: 	M4.03	A
XIV	OR Describe the applications of PERT.	M4.02	U

Question Wise Analysis

Course : Production & Operations Management

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.04	R	1	1 min
I.2	M 1.06	R	1	1 min
I.3	M 2.06	R	1	1 min
I.4	M 2.02	R	1	1 min

I.5	M 2.03	R	1	1 min
I.6	M 3.09	R	1	1 min
I.7	M 3.03	R	1	1 min
I.8	M 4.03	R	1	1 min
I.9	M 4.06	R	1	1 min
II.1	M 1.04	U	3	6 min
II.2	M 1.06	U	3	6 min
II.3	M 2.01	U	3	6 min
II.4	M 2.05	U	3	6 min
II.5	M 2.09	U	3	6 min
II.6	M 3.04	U	3	6 min
II.7	M 3.06	U	3	6 min
II.8	M 4.01	R	3	6 min
II.9	M 4.05	U	3	6 min
II.10	M 4.03	A	3	6 min
III	M 1.02	U	7	20 min
IV	M 1.08, 1.09	U	7	20 min
V	M 2.08	U	7	20 min
VI	M 2.03	U	7	20 min
VII	M3.03	U	7	20 min
VIII	M 3.04	U	7	20 min
IX	M 4.07	U	7	20 min
X	M4.02	A	7	20 min
XI	M 3.09	U	7	20 min
XII	M 3.07	U	7	20 min
XIII	M 4.03	A	7	20 min
XIV	M 4.02	U	7	20 min

Course : Production & Operations Management

Blue Print

Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	32	2	2	2	6	2	14	6	22
2	14	30	3	3	3	9	2	14	8	26
3	15	31	2	2	2	6	4	28	8	36
4	14	30	2	2	3	9	4	28	9	39
Total	58	123	9	9	10	30	12	84	31	123

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	16	13
Understanding	91	73
Applying	17	14
Analysing		
Evaluating		
Creating		
Total	123	100

SCHEME OF VALUATION

(Scoring Indicators)- Set I

Production and Operations Management

I.

1. Continuous production
2. Product or Line Layout
3. Statistical quality control
4. Sales forecast
5. Market test
6. Total Quality Management
7. Centralized purchasing
8. Critical path
9. Project

II.

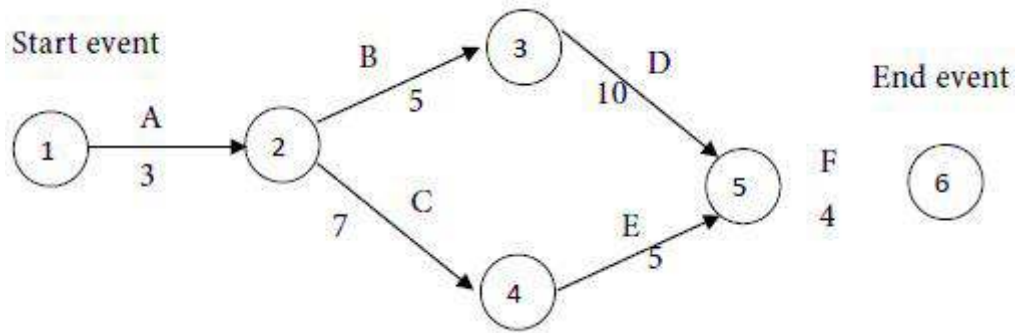
1. There are two main types of production systems : (i) Continuous System (ii) Intermittent System. Continuous or flow production refers to the manufacturing of large quantities of a single or at most a very few varieties of products with a standard set of processes and sequences. The mass production is carried on continuously for stock in anticipation of demand. Intermittent situations are those where the facilities must be flexible enough to handle a variety of products and sizes or where the basic nature of the activity imposes change of important characteristics of the input.

2. There are three main types of plant layout: 1) Functional or process layout, 2) product or line layout, and 3) stationary layout. Process Layout is also known as the functional layout and is characterized by keeping similar machines or similar operations at one location (place). Product Layout implies that various operations on a product are performed in a sequence and the machines are placed along the product flow line i.e. machines are arranged in the sequence in which a given product will operate upon. The manufacturing operations require the movements of men, machines, and materials, in the product layout and process layout generally the machines are fixed installations and the operators are static in terms of their specified work stations.
3. Production planning and control is a predetermined process that plans, manages and controls the allocation of human resource, raw material, and machinery to achieve maximum efficiency. Production planning is a sequence of steps that empower manufacturers to work smarter and optimize their production process in the best possible manner. This helps manufacturers work smarter by efficiently managing internal resources to meet customer demand.
4. To establish the desired quality standards which are acceptable to the customers, to discover flaws or variations in the raw materials and the manufacturing processes in order to ensure smooth and uninterrupted production, to evaluate the methods and processes of production and suggest further improvements in their functioning, to study and determine the extent of quality deviation in a product during the manufacturing process, to analyse in detail the causes responsible for such deviation, to undertake such steps which are helpful in achieving the desired quality of the product.
5. Aggregate planning is the process of developing, analyzing, and maintaining a preliminary, approximate schedule of the overall operations of an organization. The aggregate plan generally contains targeted sales forecasts, production levels, inventory levels, and customer backlogs.
6. Centralized purchasing is the control by one headquarters department of all purchasing that is undertaken by a business. This allows for central management and volume purchases that lead to better prices and terms as well as the ability to work with larger suppliers.
7. Value analysis is an approach to improving the value of a product or process by understanding its constituent components and their associated costs. It then seeks to find improvements to the components by either reducing their cost or increasing the value of the functions.
8. (a) To minimize idle resources.
(b) To minimize the total project cost.

(c) To trade off between time and cost of project.

(d) To minimize production delays, interruptions and conflicts.

(e) To minimize the total project duration.
9. Projects have several characteristics:
 - Projects are unique.
 - Projects are temporary in nature and have a definite beginning and ending date.
 - Projects are completed when the project goals are achieved or it's determined the project is no longer viable.



III.

BASIS FOR COMPARISON	PRODUCTION MANAGEMENT	OPERATIONS MANAGEMENT
Meaning	Production Management connotes the administration of the range of activities belonging to the creation of products.	Operations Management refers to the part of management concerned with the production and delivery of goods and services.
Decision Making	Related to the aspects of production.	Related to the regular business activities.
Found in	Enterprises where production is undertaken.	Banks, Hospitals, Companies including production companies, Agencies etc.
Objectives	To produce right quality goods in right quantity at right time and at least cost.	To utilize resources, to the extent possible so as to satisfy customer wants.

IV. Process Charts are one of the simpler forms of workflow charting. A variety of process charts has been designed to meet the needs of a particular level or stage of analysis; they can be used at a detailed level (recording activity at a specific work station or workplace), but also at the wider system, process or procedure level. There are three types of process charts and they are: i) Outline process chart ii) Two handed process chart iii) Flow process chart.

i) Outline Process Chart:

An outline process chart is the first step to a detailed analysis of a process. It gives the overall view of the whole process by recording in sequence only the main operations and inspections. It uses only operations and inspection symbols.

ii) Two Handed Process Chart:

The two handed process chart gives a synchronised and graphical representation of the sequence of manual activities of the worker. It is called two handed process chart since it records the activities of the left hand and the right hand of the worker as related to each other. A time scale can also be provided on the chart. This chart is used for repetitive works of short duration.

iii) Flow Process Chart:

A flow process chart records all the events in sequence using process chart symbols and marks distance and time taken for completing an activity.

- V. Capacity is defined as the ability to achieve, store or produce. For an organization, capacity would be the ability of a given system to produce output within the specific time period. In operations, management capacity is referred as an amount of the input resources available to produce relative output over period of time. In general, terms capacity is referred as maximum production capacity, which can be attained within a normal working schedule.

Capacity planning is essential to determine optimum utilization of resource and plays an important role decision-making process, for example, extension of existing operations, modification to product lines, starting new products, etc.

The ultimate goal of capacity planning is to meet the current and future level of the requirement at a minimal wastage. The three types of capacity planning based on goal are lead capacity planning, lag strategy planning and match strategy planning.

Effective capacity planning is dependent upon factors like production facility (layout, design, and location), product line or matrix, production technology, human capital (job design, compensation), operational structure (scheduling, quality assurance) and external structure (policy, safety regulations)

- VI. The sales forecast is an anticipation of sales, in monetary or physical unit for a defined period of time, in a specified marketing circumstances and geographical extent under a particular set of marketing programme. E.W. Cundiff and R.R. Still define sales forecast as an estimate of sales during a specified future period, which estimates is tied to a proposed marketing plan and which assures a particular set of uncontrollable and competitive forces.

There are three basic approaches to sales forecasting: the opinion approach which is based on experts judgements; the historical approach, which is based on past experience and knowledge; and the market testing approach, which is based on testing market through survey and research. No one technique of sales forecasting can be applied to all organizations, nor all factors that set a sales forecast be supplied by one individual or statistical department.

Depending upon the type of problem involved, and the degree of accuracy needed a concern selects the approach which may be more suitable to achieve the predetermined goal.

VII. There are several methods that may be utilized to request items or services. They are: blanket orders; check requests; petty cash; purchasing credit card; and purchase requisitions and orders.

Blanket Orders: An open order for a specified period which authorizes the vendor to allow items/services to be charged by one of the authorized persons listed on the order. A blanket order is for items/services usually less than \$5,000, including freight.

Check Requests: A form used to request payment for certain items. Items which are normally paid on a check request are: conference/seminar registration fees; fellowships and honorariums; inter-library loan; lodging deposits; lodging expenses for non-employees; memberships; postage; refunds/overpayments; royalty payments; agency fund expenditures (student organizations/payroll deductions) and petty cash replenishment requests.

Petty Cash: A petty cash fund is established in a department to improve the efficiency of departmental operations. These funds are available to purchase minor items when it would be otherwise impractical to purchase items through the usual purchasing methods. Petty cash is for items less than \$200.00 and not on the prohibited list.

Purchasing Credit Card: A credit card to provide departments and requisitioners with a widely accepted method of ordering small dollar, petty cash type items. The credit card is the preferred method for purchasing items less than \$5,000 and not on the prohibited list.

Purchase Requisitions and Orders: A form used to request purchase of supplies, equipment, and services.

VIII.

Centralized Purchase	Decentralized Purchase
1. Control on buying is exercised effectively	Effective control is not possible.
2. Economy in large scale purchase is possible	Large scale benefits are not available.
3. Better terms on purchase are possible.	Bargaining power is very low.
4. Skills of the purchasing officer are high.	Purchasing skill is available from the purchaser or purchasing officer.
5. Purchasing specialization is obtained.	Purchasing specialization is not obtained.

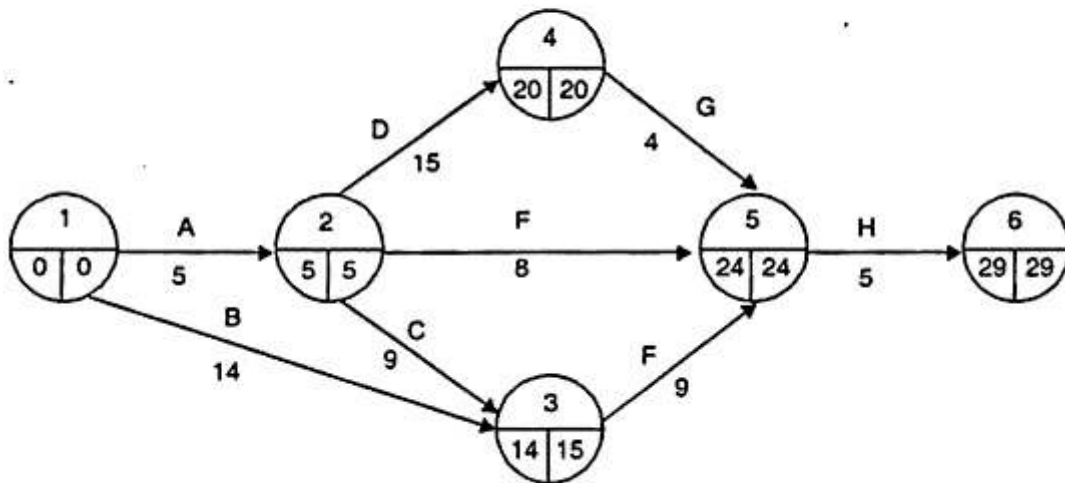
6. Uniformity in purchase is followed.	There is a lot of difference in purchase.
7. Standard materials are purchased.	Quality of the material is questionable.
8. Less finance is enough.	More finance is required.
9. Need of less storage space.	Need of more storage space.
10. It is easy to regulate purchase policy.	Regulation of purchase policy is difficult.
11. Establishment cost is high.	Establishment cost is low.
12. Delay in purchase is possible.	Purchase of materials is very quick.
13. Internal transport cost is incurred.	There is no internal transport cost.
14. Advantage of local market cannot be obtained.	Advantage of local market can be enjoyed.
15. There is a misunderstanding between production centre and purchase department.	There is no such misunderstanding since the concerned department purchases the materials.
16. There is a chance for favoritism in purchase.	There is no favoritism in purchase.
17. There is a blocking of working capital.	There is no blocking of working capital.

IX.



X. The network diagram for the given data is shown in fig. below. The earliest time and variance of each activity is computed by using the formula.

$$t_c = \frac{a + 4m + b}{6} \text{ and } \sigma^2_t = \left(\frac{b-a}{6}\right)^2$$



PERT Network diagram

(ii) Calculation activity duration and scheduling times.

Activity	Time estimates			t_c	(σ_t^2)	Earliest time		latest time	
	a	m	b			Start	finish	Start	Finish
A	2	4	12	5	25/9	0	5	0	5
B	10	12	26	14	64/9	0	14	1	15
C	8	9	10	9	1/9	5	14	6	15
D	10	15	20	15	25/9	5	20	5	20
E	7	7.5	11	8	4/9	5	13	16	24
F	9	9	9	9	0	14	23	15	24
G	3	3.5	7	4	4/9	20	24	20	24
H	5	5	5	5	0	24	29	24	29

(iii) The critical path of the project is 1-2-4-5 -6, critical activities being A, D, G and H.

The expected project length is the sum of duration of each critical activity. Expected project length = 5 + 15 + 4 + 5 = 29 weeks.

Variance project length is obtained by summing variance of each critical activity.

$$\text{Variance of project} = \frac{25}{9} + \frac{25}{9} + \frac{4}{9} + 0 = 6$$

(iv) The required probability can be determined by finding the area under the normal curve to the left of X = 30

Now, the probability of completing the project within the 30 week deadline is

$$\begin{aligned}
P(X \leq 30) &= 0.5 + P(\mu < x < 30) \\
&= 0.5 + P(0 \leq Z \leq 0.41) \\
&= 0.5 + 0.1591 \\
&= 0.6591
\end{aligned}$$

Where

$$Z = \frac{\text{Due date} - \text{Expected date}}{\sigma t}$$

$$Z = \frac{30 - 29}{\sqrt{6}} = 0.41$$

XI. There are a wide variety of techniques which can be used to help develop ideas and thinking in a programme or project management environment. Many of these tools have been around for a long time and originated in areas not directly related to PPM. Some of the tools and techniques that can be used in programme and project management are

1). SWOT - strengths, weaknesses, opportunities, threats: A SWOT analysis can be used to draw out the threats and opportunities facing a programme or project. It has the advantage of being quick to implement and is readily understood. Analysis of the strengths, weaknesses, opportunities and threats brings together the results of internal business analysis and external environmental analysis. Common and beneficial applications of SWOT are gaining a greater understanding and insight into competitors and market position.

2). RACI - responsible, accountable, consulted, informed: RACI diagram is used to describe the roles and responsibilities of the participants in a business or project activity in terms of producing predetermined deliverables.

3). Stakeholder matrix: A stakeholder matrix is used to map stakeholders in terms of their importance and potential impact on programme or project activity. Stakeholders are the individuals or groups who will be affected by an activity, programme or project.

4). Cause and effect diagram: Also known as fish-bone diagram, a cause and effect diagram can be used to represent event causes and potential impacts. It is a graphical representation of the causes of various events that lead to one or more impacts. Each diagram may possess several start points (A points) and one or more end points (B points).

5). Risk map with RAG status: This is a simple representation of risk in terms of likelihood and impact. It requires that the probability of a risk occurring is classified as low, medium or high, with a similar classification for the impact if the risk materialises.

6). Summary risk profile: A summary risk profile is a simple mechanism to increase the visibility of risks. It is a graphical representation of information normally found on an existing risk register. In some industry sectors it is referred to as a risk map.

7). Decision tree: A decision tree is a useful tool for enabling choice between several courses of action. It provides a highly effective structure within which options can be explored and possible outcomes can be investigated. It also helps to form a balanced picture of the risks and rewards associated with each possible course of action.

A decision tree is particularly useful when choosing between different strategies, projects or investment opportunities, particularly when resources are scarce.

8). Radar chart: Also known as a spider chart, a radar chart is a diagram that is used to show the number of risks that different projects are exposed to. Initially, the data is placed in a table that is subsequently converted into a chart. In a radar chart, a point close to the centre on any axis indicates a low value and a point near the edge is a high value.

- XII. Codification is the systematic concise representation of equipment, raw material, tools, spares, suppliers etc. It is one of the functions of stores management. It is the process of representing each item by a number, letter or symbol. It describes an item in short, thus avoiding long statement every time the need to describe the item arises.

Common Methods of Codification:

There are different types of store codes designed to suit the requirement. These may be based upon the type of store item, the purpose for which items are used or on any other application of that item. The common methods of store codification are as follows:

Alphabetical Codification: In this method, letters of the alphabet are used to describe an item. Sometimes combinations of alphabets are designed to give a mnemonic meaning.

Numerical Codification: The numerical system is based on numbers, simple numbers, block numbers or dash/stroke numbers.

Mnemonic Codification: When we use letters to help memory, we call such a system a mnemonic system. This system is therefore, a special application of the alphabetical system. Alphabets are allotted to an item according to its initials or abbreviation.

Combined Alphabetical & Numerical (Alphanumeric) Codification: This system is the combination of both the alphabetical and the numerical system described before. In order to improve upon the alphabetical system, and also to take advantage of the numerical system, this system allows alphabets to be retained to a limited extent and then uses the number codes. The materials are first grouped under some main classes and then it allows sub grouping under numbering system.

Decimal Codification: Under this system of codification, within the range of ten numerals 0-9, some significance is attached to every digit in the code. Thus, the whole range of items in stores can be codified without difficulty. Generally 7 to 8 digits will be found sufficient for all practical purposes, but it may also be extended upto 10 digits in order to accommodate any other characteristic of the material desired to be described.

British/Brisch Codification: The Brisch system named after a prominent consulting engineer in the UK consists of seven digits and is applied in three phases. The items are grouped into suitable preliminary categories, such as assemblies, sub assemblies, components, off the shelf items. After these preliminary categories, items are grouped within the respective groups in order to bring similar items together.

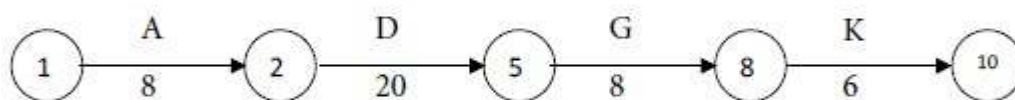
Kodak Codification: The Kodak system has been developed by East Man Kodak Company of New York which consists of 10 digits of numerical code. The logic of major grouping is based on sources of supply. All materials are divided into 100 basic classifications, contributed only by procurement considerations.

Colour Codification: This system uses colour marking for easier and more ready identification on metals, bars, pipe, drum and cable.

XIII.

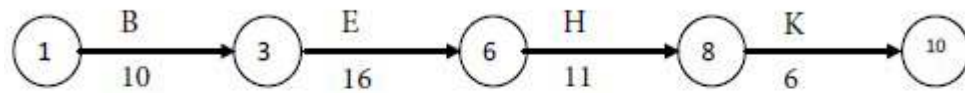
In all, we identify 4 paths, beginning with the start node of 1 and terminating at the end node of 10. They are as follows:

Path I



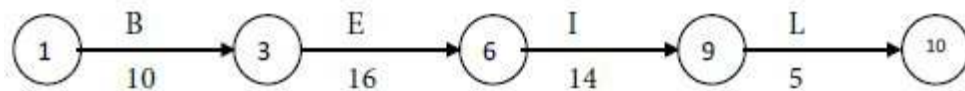
Time for the path = $8 + 20 + 8 + 6 = 42$ units of time.

Path II



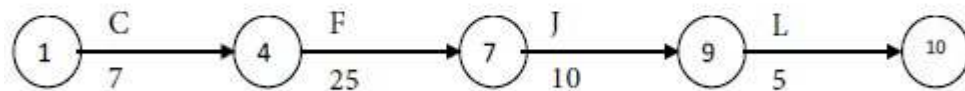
Time for the path = $10 + 16 + 11 + 6 = 43$ units of time.

Path III



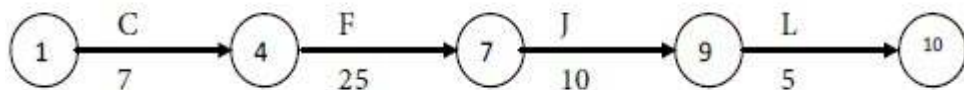
Time for the path = $10 + 16 + 14 + 5 = 45$ units of time.

Path IV



Time for the path = $7 + 25 + 10 + 5 = 47$ units of time.

Compare the times for the four paths. Maximum of $\{42, 43, 45, 47\} = 47$. We see that the following path has the maximum time and so it is the critical path:



The critical activities are C, F, J and L. The non-critical activities are A, B, D, E, G, H, I and K. The project completion time is 47 units of time.

XIV.

1. Research and Development Project (R and D Projects)
2. Equipment Maintenance and hauling
3. Construction Projects (Buildings, bridges, and Dams)

4. Setting up new industries
5. Planning and launching of new products
6. Design of plants, machines, and systems
7. Shifting the manufacturing location to another location
8. Control of production in large shops
9. Market penetration programs
10. The organization of big programs, conferences etc.

MODEL QUESTION PAPER

PRODUCTION & OPERATIONS MANAGEMENT- SET II

(Maximum marks: 75)

(Time: 3 hours)

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 marks)

1	State two differences between production and operations management.	M 1.02	R
2	The type of layout generally employed for industries engaged in job shop production and non-repetitive kind of production is	M 1.06	R
3	List out two mathematical models used for aggregate planning.	M 2.09	R
4	Recall the meaning of Statistical Quality Control.	M 2.06	R
5	Mention two advantages of decentralized purchasing.	M 3.04	R
6	Name the organized approach to identify unnecessary costs associated with any product, materials and part in manufacturing industry.	M 3.06	R
7	State the significance of Total Quality Management.	M 3.09	R
8	Mention any four objectives of network analysis.	M 4.01	R
9	Outline two differences between the project management tools, PERT and CPM.	M 4.02	R

PART B

II Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24marks)

1	A production function relates physical inputs to physical outputs. In this regard, state <i>any three</i> functions of production management.	M 1.03	R
2	Plant layout is art and science of planning, processing of product unit in effective manner. Justify the statement by citing <i>any three</i> objectives of plant layout.	M 1.05	U
3	Discuss the significance of control charts in production management.	M 1.09	U
4	Quality control ensures effective utilization of available resources and reduction in the cost of production. Do you think so? If so, justify.	M 2.05	A
5	State the concept of capacity planning.	M 2.08	U

6	Outline the significance of procurement in production.	M 3.01	R
7	Expand ISO. What other quality standards are there in ISO 9000 series and what are they used for in POM?	M 3.09	U
8	How does codification play an important role in operations of management? Write in brief.	M 3.07	U
9	Discuss any six characteristics of a project.	M 4.05	R
10	Interpret the significance of project management with respect to project completion.	M 4.06	U

PART C

Answer all questions. Each question carries seven marks

(6x7=42 marks)

III	Briefly explain the three types of production systems used in manufacturing industry.	M1.04	R
	OR		
IV	Summarize the different plant layouts used in manufacturing industry.	M1.06	R
V	Define process chart. (1) Outline the different process charts used for method study. (6)	M1.07	U
	OR		
VI	Define process planning. (1) State the significance of process planning. How is it effective in augmenting the business methods of a company? (6)	M1.09	U
VII	Define sales forecasting. (1) Brief any six methods of sales forecasting. (6)	M2.03	R
	OR		
VIII	Define production planning. (1) Brief any six advantages of production planning. (6)	M2.01	R
IX	Indicate the type of purchasing principle in the following sentences: This principle checks the suitability of items purchased for a given purpose? This principle ensures and maintains a regular flow of materials for carrying the production activity? This principle helps to fix the reorder level of material is for each item? This principle depends on quality, delivery time and ultimate life of the material, demand and supply curve, extent of competition? (2) Also, brief the above principles in few sentences. (5)	M3.05	A
	OR		
X	Copenhagen is the capital and most populated city of Denmark, with a population of almost 2 million. Its total procurement volume is €1.5bn per	M3.02	A

	year, giving it a large influence on the supply of goods and services within the Danish market. The City has set a target for 90% of food served in municipal facilities to be organic. In 2019 Copenhagen decided to also ensure that it was buying seasonal produce in order to further support efforts to strengthen biodiversity in agricultural production in both Danish and foreign producers. Extensive market engagement was undertaken in order to ensure the requirements were ambitious yet achievable, and bidders were also asked to complete 'seasonal wheels' in order to demonstrate that varieties of the most commonly used products would be available throughout the year. According to the paragraph, what do you think was the main criteria (subject matter) of the procurement process? (1) Outline the steps which you think were taken for the procurement process. (6)		
XI	State the concept of Business Process Re-engineering. (1) CASE: Conflict of Interests The GM (Works) has problems with manufacturing budgets, meeting cost reduction targets, and dealing with new products manufacturing schedules. When an in-depth interview (non-directive type) was conducted between the GM (Works) and the Chairman of the Company, the GM (Works) explained that many things are happening in the Company about which he is ignorant, particularly the preparation, new product integration, etc. He agrees to the view that the Company is interested in high-growth and high-profit, but he has never been given an opportunity to review his own scheme of things and explain to the top management. The production culture of the company has never been assessed whereas the stringent rules are being directed by the finance and personnel departments. And sometimes, show cause notices are being served to supervisors and senior employees. The Company is introducing new products without assessing the capability of the manufacturing system and the resources. (1) Under the above situation, if you are asked to work as a consultant to show the perspectives to the Board of Management, what action plans would you suggest? (2) (2) Does Business Process Re-engineering (BPR) help in situations like these? Justify. (4) OR	M3.09	A

XII	Define Total Quality Management. (1) Identify any case study in a manufacturing company like Ford, Toyota or any other and explain an instance on how they have went about implementing TQM in their company for production processes. (6)	M3.09	A																																				
XIII	Project Evaluation Review Technique, or PERT, is used to identify the time it takes to finish a particular task or activity. In the 1950s, the PERT was developed by the <u>US Navy</u> to manage the Polaris submarine missile program of their Special Projects Office. What is PERT and what are its two important advantages? (3) Discuss the parameters used to estimate project time requirements in PERT. (4) OR	M4.02	A																																				
XIV	Define Critical Path Method. (CPM) (1) Suppose you have a list of tasks as shown below. Organize all tasks into a flowchart and note their durations and help to identify the critical path to finish the task. (6) <table border="1"> <thead> <tr> <th>Task ID</th><th>Task Description</th><th>Task Predecessors</th><th>Task Duration (hours)</th></tr> </thead> <tbody> <tr> <td>A</td><td>Project start</td><td></td><td>0</td></tr> <tr> <td>B</td><td>Buy materials for A</td><td>A</td><td>10</td></tr> <tr> <td>C</td><td>Buy materials for B</td><td>A</td><td>20</td></tr> <tr> <td>D</td><td>Build A</td><td>B, C</td><td>30</td></tr> <tr> <td>E</td><td>Build B</td><td>B, C</td><td>20</td></tr> <tr> <td>F</td><td>Polish and finish B</td><td>E</td><td>40</td></tr> <tr> <td>G</td><td>Join A and B</td><td>D, F</td><td>20</td></tr> <tr> <td>H</td><td>Project finish</td><td>G</td><td>0</td></tr> </tbody> </table>	Task ID	Task Description	Task Predecessors	Task Duration (hours)	A	Project start		0	B	Buy materials for A	A	10	C	Buy materials for B	A	20	D	Build A	B, C	30	E	Build B	B, C	20	F	Polish and finish B	E	40	G	Join A and B	D, F	20	H	Project finish	G	0	M4.03	A
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H	Project finish	G	0																																				

Question Wise Analysis

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.02	R	1	1 min
I.2	M 1.06	R	1	1 min
I.3	M 2.09	R	1	1 min
I.4	M 2.06	R	1	1 min
I.5	M 3.04	R	1	1 min
I.6	M 3.06	R	1	1 min
I.7	M 3.09	R	1	1 min
I.8	M 4.01	R	1	1 min
I.9	M 4.02	R	1	1 min
II.1	M 1.03	R	3	6 min
II.2	M 1.05	U	3	6 min
II.3	M 1.09	U	3	6 min
II.4	M 2.05	A	3	6 min
II.5	M 2.08	U	3	6 min
II.6	M 3.01	R	3	6 min
II.7	M 3.09	U	3	6 min
II.8	M 3.07	U	3	6 min
II.9	M 4.05	R	3	6 min
II.10	M 4.06	U	3	6 min
III	M 1.04	R	7	20 min
IV	M 1.06	R	7	20 min
V	M 1.07	U	7	20 min
VI	M 1.09	U	7	20 min
VII	M 2.03	R	7	20 min
VIII	M 2.01	R	7	20 min
IX	M 3.05	A	7	20 min
X	M 3.02	A	7	20 min
XI	M 3.09	A	7	20 min
XII	M 3.09	A	7	20 min

XIII	M 4.02	A	7	20 min
XIV	M 4.03	A	7	20 min

Course : Production & Operations Management

Blue Print

Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	32	2	2	3	9	4	28	9	39
2	14	30	2	3	2	6	2	14	6	23
3	15	31	3	2	3	9	4	28	10	39
4	14	30	2	2	2	6	2	14	6	22
Total	58	123	9	9	10	30	12	84	31	123

Blue Print

**Cognitive Level Mark
Distribution**

Cognitive Level	Marks	% of Marks
Remembering	46	37
Understanding	32	26
Applying	45	37
Analysing		
Evaluating		
Creating		
Total	123	100

SCHEME OF VALUATION

(Scoring Indicators)- Set II

PRODUCTION & OPERATIONS MANAGEMENT

Questi on no	Scoring indicators	Split up score	Sub total	Total
	PART A			
I(1)	• Production management deals with manufacturing / production / construction, while operation management cover both products and services with that product • Time for production management and product consumption takes time, while services by the operation management consumed immediately. • Production management needs more capital equipment including lands & factories, more investment and less labours, while services in operation management require more labour (hr) and lesser capital investment. • Production management needs no customer contact	1		

	during production, while operation management always deals with customers directly and indirectly, along with the contacts with many more 3rd parties like suppliers, dealers, communication agencies etc. (any two)		1x9	9
(2)	The type of layout generally employed for industries engaged in job shop production and non-repetitive kind of production is process layout .	1		
(3)	Mathematical model for aggregate planning (aggregate planning methods) a) Graphic method b) Transportation model c) Linear programming model d) Heuristic method (any two models required)	1		
(4)	Statistical quality control, the use of statistical methods in the monitoring and maintaining of the quality of products and services.	1		
(5)	- Materials can be purchased by each department locally as and when required. - Materials are purchased in right quantity of right quality for each department easily. - No heavy investment is required initially. - Purchase orders can be placed quickly. - The replacement of defective materials takes little time. (any two)	1		
(6)	The organized approach to identify unnecessary costs associated with any product, materials and part in manufacturing industry—Value Analysis	1		
(7)	TQM can have an important and beneficial effect on employee and organizational development. By having all employees focus on quality management and continuous improvement, companies can establish and uphold cultural values that create long-term success to both customers and the organization itself.	1		
(8)	1. Powerful tool of planning, scheduling and control. 2. Shows the inter-relationships of the activities of a project or a programme. 3. Minimises total cost where the cost of delays and cost of resources required to carry out the tasks can be measured. 4. Minimise total time where required e.g., in maintenance of production-line machinery in a factory. 5. Minimization of idle resources. 6. Minimise production delays. (any four)	1		
(9)	PERT deals with unpredictable activities, but CPM deals with predictable activities. PERT is used where the nature of the job is non-repetitive. In contrast to, CPM involves the job of repetitive nature. (any other differences)	1		

	<u>PART B</u>			
II (1)	<u>Functions of production management.</u> 1. Selection of Product and Design 2. Selection of Production Process 3. Selecting Right Production Capacity 4. Production Planning 5. Production Control 6. Quality and Cost Control 7. Inventory Control 8. Maintenance and Replacement of Machines <i>(brief description on any three functions)</i>	1x3	3	
(2)	Plant layout is a plan of an optimum arrangement of facilities including personnel, operating equipment, storage space, material handling equipment and all other supporting services along with the design of best structure to contain all these facilities. The objectives of plant layout: 1. Streamline the flow of materials through the plant. 2. Facilitate the manufacturing process. 3. Minimise materials handling cost.	1x3	3	
(3)	Significance of control charts- 1. It is a proven technique for improving productivity 2. It is an effective system in defect prevention 3. It prevents unnecessary process adjustments 4. It provides diagnostic information 5. It provide about process capability (any three)	1x 3	3	
(4)	Yes, quality control ensures effective utilization of available resources and reduction in the cost of production as it helps- 1. Encourages quality consciousness 2. Satisfaction of consumers 3. Reduction in production cost 4. Most effective utilisation of resources 5. Reduction in inspection costs 6. Increased goodwill etc. (any other valid points)	1x3	3	
(5)	Capacity planning is the process of determining the production capacity needed by an organization to meet changing demands for its products. In the context of capacity planning, design capacity is the maximum amount of work that an organization is capable of completing in a given period. Production managers are more concerned about the capacity planning for the following reasons • Sufficient capacity is required to meet the customers demand in time. • Capacity affects the cost efficiency of operations. • Capacity affects the scheduling system. • Capacity creation requires an investment. (any other relevant points from the contents)	1x3	3	

(6)	Procurement involves the process of selecting vendors, establishing payment terms, strategic vetting, selection, the negotiation of contracts and actual purchasing of goods. Procurement is concerned with acquiring (procuring) all of the goods, services and work that is vital to an organization. Procurement is, essentially, the comprehensive or umbrella term within which purchasing can be found.	1x3	3	24
(7)	ISO -International Organisation of Standardization 1. ISO 9000 — Quality management and Quality assurance standards 2. ISO 9001 — Quality systems: Quality in design 3. ISO 9002 — Quality systems: Production and Installation 4. ISO 9003 — Quality systems: Final inspection and test 5. ISO 9004 — Quality management and systems	1 2	3	
(8)	Codification is a process of representing each item by a number, the digit of which indicates the group, the sub-group, the type and the dimension of the item. Many organizations in the public and private sectors, railways have their own system of codification, varying from eight to thirteen digits. The first two digits represents the major groups, such as raw materials, spare parts, sub-contracted items, hardware items, packing material, tools, oil, stationery etc. The next two digits indicate the sub-groups, such as, ferrous, non-ferrous etc. Dimensional characteristics of length, width, head diameter etc. constitute further three digits and the last digit is reserved for minor variations. Codification should be compact, concise, consistent and flexible enough to accommodate new items	1x3	3 3	
(9)	Characteristics of a Project 1. Temporary 2. Definite Beginning and Completion 3. Definite Objective/Scope and Unique 4. Defined Time and Resources 5. Multiple Talents 6. Projects consist of several activities (with explanation)	0.5x6	3	
(10)	Project management is the discipline of planning, organizing and managing resources to bring about the successful completion of specific project goals and objectives. Project management aims to ensure the effective use of resources and delivery of the project objectives on time and within cost constraints. The purpose of project management is to achieve successful project completion with the resources available. A successful project is one which: • Has been finished on time • Is within its cost budget • Performs to a technical/performance standard which satisfies the end user. (any other relevant points)	1x3		

III	Three types of production systems: - 1. Continuous production refers to the manufacturing of large volume of a single or a very few varieties of products with a standard set of processes. The mass production is carried on continuously for stock in anticipation of demand. 2. Intermittent Production System- the goods are generally produced to fulfill customers' orders rather than producing against stock. Intermittent situations are those where the facilities must be flexible enough to handle a variety of products and sizes. The flow of material is intermittent 3. Batch Production- items are processed in lots or batches and a new batch is undertaken for production only when the production on all items of a bath is complete. (Name of types – 1 mark Description of the characteristics and valid points-6 marks)	1 6	7	
IV	Types of plant layouts: 1) Process layout: It is also known as functional layout. Here similar machines and services located together Ex. All the lathe machines will be at one place and all milling machines at another place and so on. 2) Product layout: It is also known as line (type) layout. The flow of product will smooth and logical. When the machines and auxiliary services are located according to the processing sequence, we prefer this layout. 3) Group layout: It is the combination of both process and product layout. In this type of layout, a set of machinery or equipment is grouped together in a section so that each group of machines or equipment is used to perform similar operations to produce a family of components. These machines grouped in to cells. 4) Fixed position layout: It is also called static product layout in which the physical characteristics of the product dictate as to which type of machine and men are brought to the product 5) Combination layout -A combination layout combines the advantages of both types of product and process layouts. (Name of layout-1 mark Explanation- 5 marks Illustration of layouts- 1mark)	1 5 1	7	

V	a) Process chart- a chart on which are graphically shown in sequence the separate details that make up a complete process (as of a particular job operation) b) Different process charts used for method study 1. Outline process chart 2. Operation process chart 3. Flow process chart 4. Two handed process chart 5. Multiple Activity Chart 6. The man machine chart or worker-machine chart 7. Flow diagram 8. String diagram 9. SIMO chart (explanation of any six of these)	1 1x6	7	
VI	a) Process planning- Process planning involves the definition, documentation, review and improvement of steps in business processes used at a company. b) It is effective in augmenting the business methods of a company(significance)- • Process planning is planned to renovate design specification into manufacturing instructions and to make products within the function and quality specification at the least possible costs. • This will result in reduced costs, due to fewer staff required to complete the same process, higher competence, by eradicating process steps such as loops and bottlenecks, • Greater precision, by including checkpoints and success measures to make sure process steps are completed precisely • Better understanding by all employees to fulfil their department objectives. • Process planning deals with the selection of the processes and the determination of conditions of the processes. • The particular operations and conditions have to be realized in order to change raw material into a specified shape. All the specifications and conditions of operations are included in the process plan. Consequently, the planning activities have immense importance for competitive advantage. (any other valid points)	1 1x6	7	
	a) Sales forecasting- It is the prediction of sales volume that a company can estimate to achieve in specified period of time in future. In the words of Philip Kotler “The company sales forecast is the expected level of company sales based on a chosen marketing plan and assumed	1	7	

	The right quantity is the quantity that may be purchased at a time with the minimum total cost and which obviates shortage of materials. Ensuring and maintaining a regular flow of materials for carrying the production activity is the vital aim of any purchase organisation. 3. Right Time: In case of items used regularly, right time means the time when the stock reaches the minimum level. The reorder level of material is fixed for each item under the principle of right time. 5. Right Price: It is the main object of any organisation to procure the material items at the right price. It is that price which brings the best ultimate value of the money invested in purchasing the materials. • (Writing down the 5 principles in any random order- 1 mark) • (explanation of the correct purchasing principles- 4 marks)	5		
X	(a) the main criteria (subject matter) of the procurement process is provision of procuring 100% organic, seasonal fruit and vegetables. (b) Procurement process- Steps: - 1. Recognition of the need: The initiation of procedure starts with the recognition of the need by the needy section. 2. Selection of the supplier: The process of selection of supplier involves two basic aspects: searching for all possible sources and short listing out of the identified sources. 3. Placing the order: Once the supplier is selected the next step is to place the purchase order. Purchase order is a letter sent to the supplier asking to supply the said material. 4. Follow-up of the order: Follow-up procedure should be employed wherever the costs and risks resulting from the delayed deliveries of materials are greater than the cost of follow-up procedure. 5. Receiving and inspection of the materials and payment of the invoice: The receiving department receives the materials supplied by the vendor. The quantity is verified and tallied with the purchase order. When the goods are received in satisfactory condition, the invoice is checked before it is approved for the payment. 6. Maintenance of the records and vendor relations: In the industrial firms, most of the purchases are repeat orders and hence the past records serve as a good guide for the future action. Also, the quantum and	2	1x6	7

	frequency of the transactions with the same key suppliers provide a platform for the purchase department to establish and maintain good relations with them (answers relating to the paragraph in his/her viewpoint but should follow the above said steps.)			
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XIII	entire organization needs to operate as a single unit in the pursuit of excellence. b) Case study Example- TOYOTA -The Toyota way (kaizen) aims at integrating the workforce suggestions while eliminating overproduction and manufacturing wastes helps the company to respect all the stakeholders and give clients first priority. • The company extended the management responsibility past the instantaneous services and products • Toyota examined how consumers applied the products generated and this enabled the company to develop and the improve its commodities • Toyota focused on the insubstantial impacts on the procedures as well as how such effects could be minimized through optimization • Toyota focused on the kaizen (incessant processes development) in order to ensure that all procedures are measurable, repeatable, and visible. (Introducing any company- 2 marks Main aspects to check is in the case study there should be mention of either BPR, Kaizen, ISO series or TQM tools, carries-4 marks) (a) PERT -Program Evaluation and Review Technique PERT is used to schedule, organize, and coordinate tasks within a project. It is basically a method to analyze the tasks involved in completing a given project, especially the time needed to complete each task, and to identify the minimum time needed to complete the total project. The main objective of PERT is to facilitate decision making and to reduce both the time and cost required to complete a project. Two advantages – 1. It helps maximize the use of resources. 2. It makes project planning more manageable. 3. It's useful even if there is little or no previous schedule data. 4. It enables project managers to better estimate or determine a more definite completion date. (Any two) (b) Parameters used to estimate project time requirements in PERT: • Optimistic time – The least amount of time it can take to complete a task	2+4	7	
		1	7	
		2		
		4		

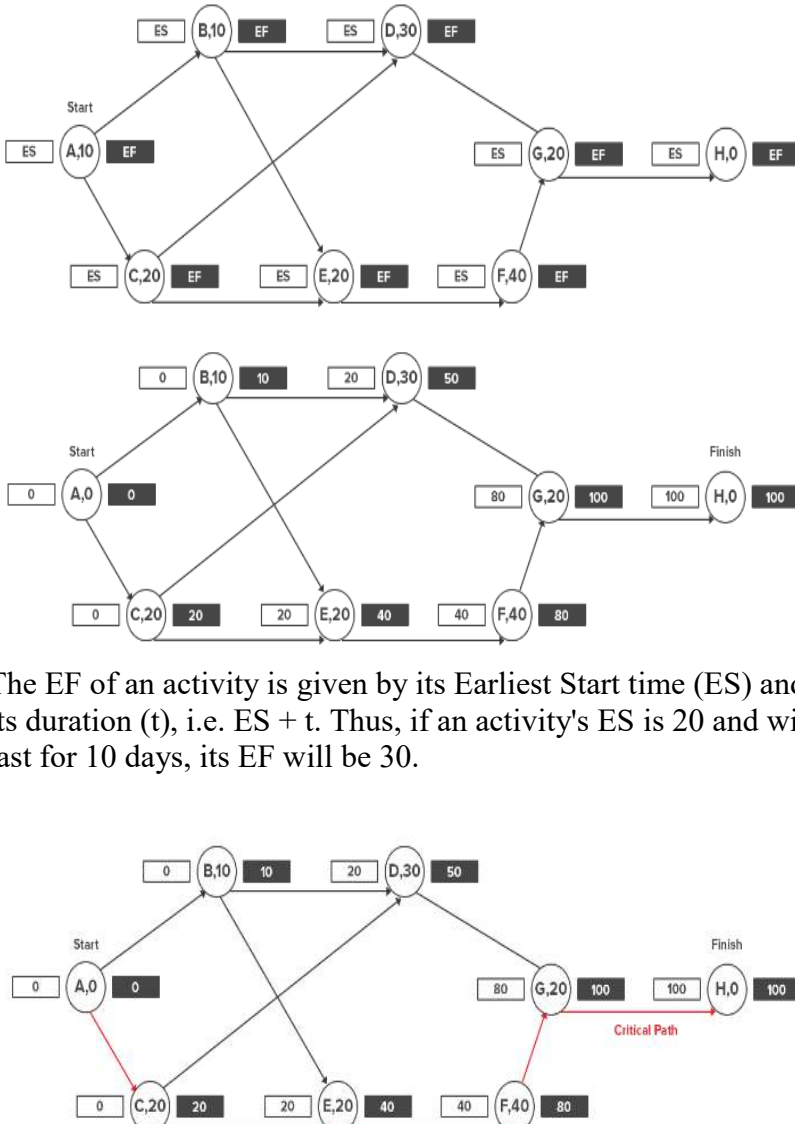
XIV

- Pessimistic time – The maximum amount of time it should take to complete a task
- Most likely time – Assuming there are no problems, the best or most reasonable estimate of how long it should take to complete a task.
- Expected time – Assuming there are problems, the best estimate of how much time will be required to complete a task.

(a) Critical Path Method. (CPM)-The Critical Path Method is the sequence of scheduled activities that determines the duration of the project.

(b) To identify the critical path to finish the task.

Task ID	Task Description	Task Predecessors	Task Duration (hours)
A	Project start		0
B	Buy materials for A	A	10
C	Buy materials for B	A	20
D	Build A	B, C	30
E	Build B	B, C	20
F	Polish and finish B	E	40
G	Join A and B	D, F	20
H	Project finish	G	0

XIV	Mark the Earliest Start (ES) time to the left of the activity, and the Earliest Finish (EF) time to the right. Mark the Start Time (S) to the left and right of the first activity. Usually, this would be 0. Mark the Earliest Start (ES) time of each activity. This is given by the largest number to the right of the activity's immediate predecessor (i.e. its Earliest Finish time, or EF). If the activity has two predecessors, the one with the later EF time would give you the ES of the activity.  The EF of an activity is given by its Earliest Start time (ES) and its duration (t), i.e. $ES + t$. Thus, if an activity's ES is 20 and will last for 10 days, its EF will be 30. The longest path will be the “critical path”.	1	7
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